NTNU
Kunnskap for en bedre verden

IT3708 - Bio-Inspired Artificial Intelligence

Project 2

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1 Introduction

Efficient scheduling in home-care services is crucial for maximizing patient care while minimizing travel time for nurses. This project focuses on a simplified Home-Care Vehicle Routing Problem (VRP), where nurses must visit patients within specific time windows and capacity limits, ensuring each patient is visited exactly once. The goal is to minimize total travel time, allowing nurses to spend more time providing care rather than traveling. The problem is inspired by real-world solutions from Visma Resolve, which develops meta-heuristic solvers for home-care logistics in Norwegian municipalities.

To solve this NP-hard problem, we implement a multi-island genetic algorithm (GA) with specialized operators. Each island evolves a population of candidate routes using a combination of mixed crossovers, hybrid mutations, and deterministic crowding to maintain diversity. Routes are evaluated by the Prins algorithm, which calculates travel feasibility within time windows and depot constraints. Periodic migrations between islands and entropy-based population injections further enhance exploration. This approach allows us to efficiently generate high-quality solutions while illustrating practical applications of meta-heuristics to complex, real-world scheduling problems in healthcare.

2 Experimental setup

2.1 Chromosome Representation

Each individual is represented as a *giant-tour permutation* of patient IDs:

$$\texttt{genotype} = [\pi_1, \pi_2, \ldots, \pi_n],$$

where each patient appears exactly once [1]. In addition, each individual stores a `splits` vector with cumulative end-indices that partition the giant tour into nurse routes. For example, if `splits` is $[4, 9, 12]$, then route 1 is patients 1:4, route 2 is 5:9, and route 3 is 10:12.

This representation separates:

1. Ordering decisions (handled by GA operators on permutations), and
2. Route partitioning decisions (handled by the split procedure, see Section 2.2).

2.2 Local Search / Split algorithm

The Prins split algorithm is used to evaluate a permutation by optimally partitioning it into feasible routes [1], [2]. Given an individual's giant-tour order, the split algorithm (SPA) determines the minimum-travel-time segmentation of the giant tour into routes while enforcing: 1) nurse capacity, 2) patient time windows (including waiting), and 3) return-to-depot time.

Importantly, the split algorithm may in principle generate more routes (i.e., require more nurses) than are available. This means that the "number of nurses" constraint is in practice not enforced by the algorithm. In such cases, the algorithm treats the solution as feasible, even though it is considered infeasible in practice.

The fitness function is simply defined as the total travel time of the proposed solution, as the goal of the project as a whole is to minimize total travel time.

Thus, while the GA searches over permutations, the split algorithm computes the best feasible segmentation of the permutation and assigns its fitness accordingly.

2.3 Population

For population management an island model is used with the parameters shown in table 1.

Table 1: Population parameters

Parameter	Value
Island Count	5
Island Pop. Size	600
Max Generations	10000
Migration Interval	100
Migration Count	2

Each island evolves independently throughout the run. To share genetic information, migration is performed every 100 generations using a ring topology: each island sends its best 2 individuals to the adjacent island, replacing the worst-performing individuals there. This "ring" setup is a common way to balance independent searching with the need to occasionally share good traits, as suggested by Eiben [3].

Furthermore, to counter premature convergence, each island continuously monitors its genetic entropy. If the local entropy drops below 5%, a partial reinitialization is triggered, where the bottom 90% of the population is filled with random immigrants while the top 10% of elite solutions are preserved. This mechanism draws from the concepts of diversity maintenance in dynamic environments presented by Eiben [3].

2.4 Crossover

We employ a mixed crossover operator, applied with a probability of $P = 0.9$, designed to balance structural exploitation with global sequence exploration. The operator randomly selects between two distinct reproduction mechanisms:

Route Crossover is chosen with a probability of $P = 0.6$. It takes one complete, successful route from the first parent and puts it into the child, then fills in the remaining slots using the patient-order from the second parent.

Order Crossover is chosen with a probability of $P = 0.4$. It copies a random segment of patients from one parent and fills in the remaining ones using the order from the second parent to make sure the solution is complete.

2.5 Mutation

Table 2: Table 2: Mutation parameters

Parameter	Value
overall mutation rate	0.25
swap mutation rate	0.2
inversion mutation rate	0.6
insertion mutation rate	0.2

Mutation is applied to each individual with an overall probability shown in Table 2. If the individual is mutated, the secondary mutation probability in Table 2 decides whether to mutate using swap mutation, inversion mutation or insertion mutation.

The three mutation operators were implemented following the description in the curriculum [3]:

Swap mutation It performs random pairwise swaps between genes (with per-gene swap probability $1/n$, where n is the size of the chromosome).

Inversion mutation It inverts a random short segment of the chromosome (up to 20% of chromosome length).

Insertion mutation Removes one gene from the chromosome and insert it at a new random position.

2.6 Survival Selection

Deterministic crowding is used for survival selection. For each parent pair and offspring pair the edge-based genotype distance is computed, matching each offspring to its most similar parent. Because this is deterministic, the fitter individual of each parent/offspring pair survives. Handling it this way promotes preservation of niches by ensuring the offspring mainly compete locally with their parents, maintaining diversity while still applying some fitness pressure.

3 Results

3.1 Train Dataset

The results for fitness and percentage away from benchmark for the training instances are shown in table 3, and for the test instances in table 4.

Table 3: Results for each training instance.

Instance	Best Fitness	From Benchmark (%)
train_0	828.94	0.20
train_1	591.56	0.42
train_2	1279.15	1.68
train_3	1163.03	2.71
train_4	1279.29	1.39
train_5	1117.73	2.33
train_6	946.64	2.48
train_7	884.17	1.54
train_8	745.01	1.87
train_9	884.74	3.5

As seen in tables 3 and 4 the model manages to be within 5% of the benchmark for all instances.

3.2 Test Dataset

The results for the test instances is outlined in Table 4. We managed an average score of -0.1% from the benchmark, with both Test Instance 2 & 3 surpassing the banchmark score. Note that Test Instance 3 required more generations than the other two, and the results for instance 3 is from a run with 25k generations.

The detailed results are provided in appendices A, B and C, as well as in the project zip folder.

Table 4: Results for each test instance.

Instance	Best Fitness	From Benchmark (%)
test_instance_1	1521.14	0.55%
test_instance_2	1587.97	-0.31%
test_instance_3	2303.96	-0.55%

Figure 1 shows the route solutions for Test Instance 2, while Figure 2 illustrate the convergence of both fitness and population entropy for the same run.

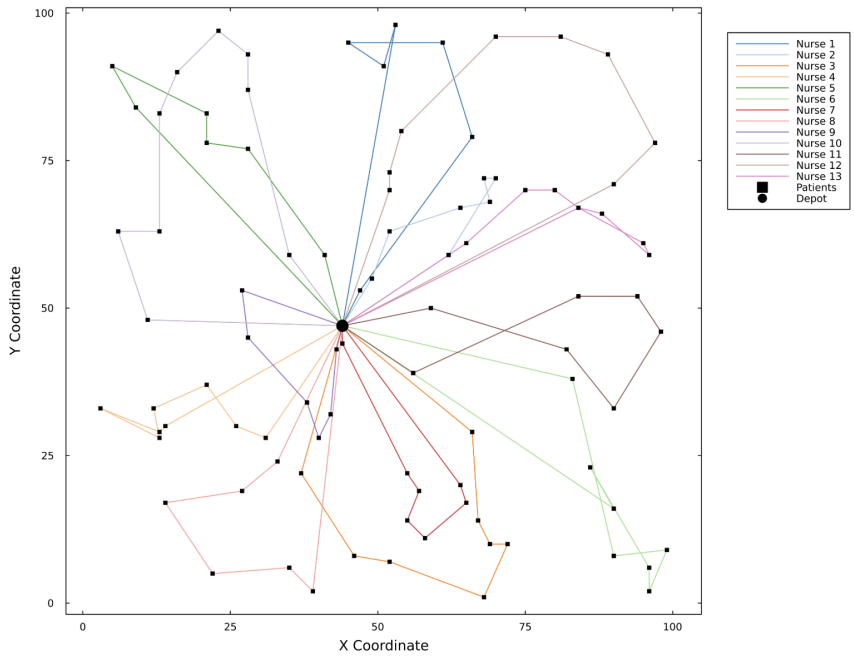

Figure 1: Route Solutions for Test Instance 2

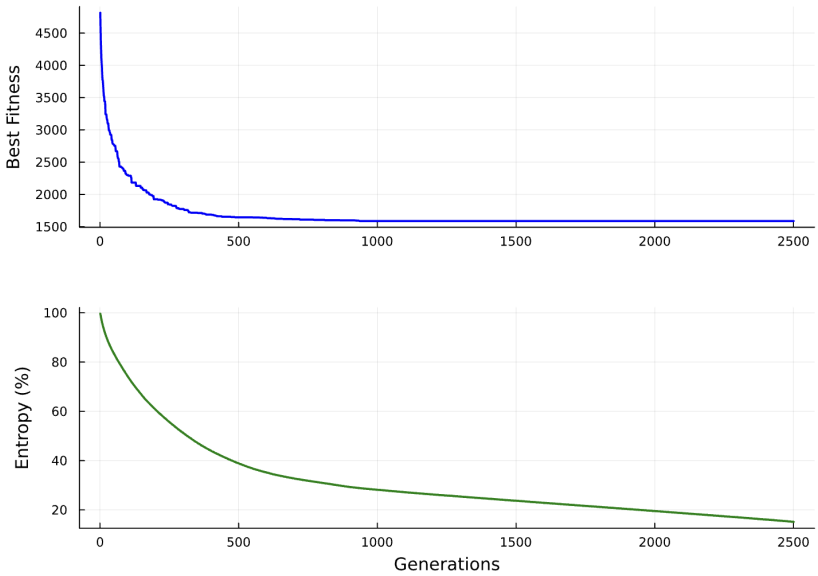

Figure 2: Convergence Plot for Test Instance 2

4 Discussion

4.1 Islands vs. One Population

Our initial evolutionary algorithm (EA) used a standard one population model where one population is evolved until convergence. However, we noticed that the diversity of the population fell relatively quickly, which increased the risk of premature convergence. The final solution leverages the island model. In this model the total population is divided into different islands that evolve independently of each other. This independence helps the EA preserve global diversity to explore and refine different route configurations locally. The two best individuals from each island are periodically migrated over to other islands to introduce new information. This migration is explorative by introducing new genetic material but still allows each island to exploit the good solutions locally.

4.2 Double selection pressure

Initially we tried a setup with both tournament selection and deterministic crowding, but this did not work as well as we hoped. The problem was double selection pressure, where the best individuals from one generation would be duplicated multiple times into the next. This made the diversity of the population drop rapidly, and the algorithm converged prematurely to a suboptimal solution.

The solution to this problem was to remove tournament selection and keep deterministic crowding. This way, selection pressure is only applied once, and the diversity of the population is kept higher for much longer. Instead of using tournament selection, we chose to use random parent selection. This ties right into the method outlined by Eiben [3], giving the algorithm the breathing room it needs to thoroughly explore and find higher-quality solutions.

4.3 Multiple mutation strategies

Instead of relying on a single mutation operator, we used a multiple mutation strategy to ensure the algorithm thoroughly explores the search space. A single mutation type can limit the kind of new routes the algorithm can discover, making it highly vulnerable to getting trapped in local optima. By randomly selecting from a variety of distinct mutation techniques, we give the algorithm the flexibility it needs to easily escape local traps and push toward the global optimum.

To achieve this, we combined three distinct operators, each serving a specific evolutionary purpose:

- Swap: We included swap mutation to handle localized fine-tuning. It is highly effective at making small adjustments between two points without tearing apart the broader sequence, allowing the algorithm to polish and optimize already strong routes.
- Insertion: Insertion is particularly valuable for correcting single misplaced patients. If one patient is severely dragging down a route's fitness, this operator allows the algorithm to extract and reposition that specific node into a more efficient slot without disrupting the relative order of the remaining schedule.
- Inversion: Finally, inversion is critical for resolving routing inefficiencies. In spatial problems, reversing a sequence of visits is often the exact move needed to untangle crossing paths and overlapping loops, immediately reducing the total travel distance.

4.4 Worth mentioning

4.4.1 Enforcing max nurses

One aspect of our implementation is that the solution representation does not explicitly enforce the constraint on the maximum number of available nurses. As described in Section 2.2, the split algorithm may in theory produce solutions that utilize more routes (and therefore more nurses) than allowed.

In practice, however, this did not occur for the instances we evaluated. For the training instances, the algorithm consistently produced solutions using roughly half of the available nurses. This behavior can be explained by the objective function: since each route must start and end at the depot, introducing additional routes generally increases total travel time (often more than including the patient in another route), making such solutions less competitive.

For the test instances, we observed somewhat higher numbers of nurses being used, but still within the available limits. While a real-world implementation should include an explicit safeguard or penalty to guarantee that the nurse limit is never exceeded, this simplification did not negatively affect solution feasibility or quality in our experimental setting.

4.4.2 Features not useful in the test set

Section 2.3 describes a safeguard that triggers a partial population reinitialization when the genetic entropy of an island drops below a predefined threshold. While this mechanism was activated for about half the training instances, it did not trigger for any of the test instances. This indicates that the population maintained a higher level of diversity throughout those runs.

A possible explanation is that the test instances produce a broader range of similarly competitive solutions. One reason for this may be differences in the spatial distribution of patients, such as less clustered layouts, which could allow for more varied route structures with comparable fitness values. As a result, the population diversity remained sufficiently high and the entropy safeguard was never activated.

5 Conclusion

This project successfully demonstrated the effectiveness of a multi-island genetic algorithm in solving the simplified Home-Care Vehicle Routing Problem. By minimizing total travel time, the algorithm optimizes nurse schedules to prioritize patient care.

The transition from a standard single-population model to a multi-island architecture proved crucial for preserving global diversity and avoiding premature convergence. Furthermore, replacing tournament selection with deterministic crowding successfully mitigated issues with double selection pressure, allowing the algorithm more breathing room to explore higher-quality solutions. The strategic use of multiple distinct mutation techniques gave the algorithm the flexibility needed to escape local optima and efficiently refine routing sequences.

Ultimately, this approach yielded highly competitive results, remaining within 5% of the benchmark across all training instances. On the test dataset, the model achieved an average distance of -0.1% from the benchmark, due to surpassing it on both Test instance 2 and 3.

6 AI Declaration

Multiple LLMs such as ChatGPT and Google Gemini has been used in the report as an aid for rephrasing paragraphs and better wording, generating a better flow.

References

[1] Jingyi Zhao et al. 'A hybrid genetic search and dynamic programming-based split algorithm for the multi-trip time-dependent vehicle routing problem'. In: *European Journal of Operational Research* 317.3 (2024), pp. 921–935. ISSN: 0377-2217. DOI: https://doi.org/10.1016/j.ejor.2024.04.011. URL: https://www.sciencedirect.com/science/article/pii/S0377221724002923.

[2] Christian Prins. 'A simple and effective evolutionary algorithm for the vehicle routing problem'. In: *Computers & Operations Research* 31.12 (2004), pp. 1985–2002. ISSN: 0305-0548. DOI: https://doi.org/10.1016/S0305-0548(03)00158-8. URL: https://www.sciencedirect.com/science/article/pii/S0305054803001588.

[3] A. E. Eiben and J. E. Smith. *Introduction to Evolutionary Computing*. 2nd. Springer, 2015. pages 69 - 70 (mutation for permutation) & pages 93 - 97 (island GA & crowding) & pages 190 - 192 (random immigrants).

A Test Instance 1 Solution

```
Nurse capacity: 205
Depot return time: 1204
```

```
Nurse 1 141.42  193     D(0) -> 12(236.0-309.0) [236-354] -> 6(321.0-407.0) [321-795] -> 15(433.63-510.63) [367-720]
    -> 81(643.0-748.0) [643-1166] -> 79(765.46-881.46) [567-904] -> 3(888.74-954.74) [60-1215] ->
    71(981.67-1087.67) [757-1171] -> D (1092.67)
Nurse 2 134.52  199     D(0) -> 48(61.0-140.0) [61-354] -> 29(155.81-262.81) [152-324] -> 44(265.81-358.81)
    [221-1379] -> 34(370.0-435.0) [370-678] -> 45(447.04-542.04) [306-1259] -> 19(614.0-729.0) [614-764] ->
    86(756.17-855.17) [408-930] -> 73(861.25-954.25) [777-1158] -> 21(966.25-1056.25) [250-1117] -> D (1080.6)
Nurse 3 155.49  201     D(0) -> 42(53.04-169.04) [46-522] -> 50(345.0-451.0) [345-1305] -> 8(648.0-735.0) [648-1807]
    -> 24(766.0-839.0) [766-1588] -> 69(855.28-931.28) [628-1030] -> 88(942.28-1042.28) [765-1769] ->
    90(1045.88-1133.88) [515-1491] -> D (1184.09)
Nurse 4 113.17  180     D(0) -> 89(26.68-90.68) [25-146] -> 64(132.0-211.0) [132-440] -> 17(529.0-593.0) [529-1194]
    -> 27(598.39-699.39) [198-1298] -> 53(776.0-891.0) [776-1326] -> 5(892.41-979.41) [487-1161] ->
    4(992.45-1104.45) [682-1391] -> D (1129.45)
Nurse 5 154.95  140     D(0) -> 83(53.25-159.25) [39-168] -> 49(163.73-272.73) [39-353] -> 36(285.76-356.76)
    [76-568] -> 82(676.0-772.0) [676-1412] -> 80(784.0-860.0) [372-946] -> 87(868.25-949.25) [791-1523] ->
    41(973.43-1065.43) [488-1471] -> 11(1087.27-1166.27) [647-1637] -> D (1176.57)
Nurse 6 117.96  202     D(0) -> 23(182.0-248.0) [182-538] -> 46(258.0-359.0) [258-574] -> 84(397.01-473.01) [71-572]
    -> 57(512.0-611.0) [512-638] -> 68(618.28-694.28) [530-982] -> 2(702.82-794.82) [521-1264] -> 93(806.82-904.82)
    [648-1742] -> 26(913.77-982.77) [662-1127] -> D (1006.81)
Nurse 7 73.54   199     D(0) -> 20(274.0-351.0) [274-374] -> 30(375.41-478.41) [148-546] -> 75(483.51-549.51)
    [432-743] -> 63(644.0-707.0) [644-1621] -> 10(731.35-839.35) [486-1406] -> 70(845.01-943.01) [285-992] -> D
    (950.08)
Nurse 8 135.84  195     D(0) -> 39(272.0-351.0) [272-1124] -> 59(365.32-478.32) [365-601] -> 35(486.38-569.38)
    [223-1015] -> 55(572.38-645.38) [279-663] -> 72(663.82-762.82) [509-1342] -> 32(774.0-863.0) [740-1841] ->
    58(865.83-961.83) [744-1035] -> 67(964.66-1078.66) [725-1127] -> D (1126.92)
Nurse 9 21.88   61      D(0) -> 40(7.28-105.28) [4-527] -> 37(796.0-876.0) [796-901] -> 14(886.77-975.77) [668-1359]
    -> D (978.6)
Nurse 10        99.98   188     D(0) -> 74(340.0-435.0) [340-713] -> 47(442.07-535.07) [440-683] ->
    76(544.29-634.29) [460-912] -> 28(640.37-702.37) [478-712] -> 9(712.67-828.67) [660-1542] -> 1(835.74-917.74)
    [353-1119] -> 31(925.74-1038.74) [762-1792] -> D (1070.76)
```

```
Nurse 11        152.92  192     D(0) -> 51(68.0-160.0) [68-379] -> 61(203.0-278.0) [203-977] -> 78(285.81-349.81)
    [181-352] -> 62(369.53-461.53) [182-834] -> 56(465.66-572.66) [410-1477] -> 33(583.84-675.84) [188-1015] ->
    7(680.31-773.31) [633-786] -> 66(789.59-876.59) [464-1504] -> D (902.66)
Nurse 12        96.04   190     D(0) -> 16(271.0-345.0) [271-522] -> 85(372.0-464.0) [372-669] -> 54(639.0-754.0)
    [639-1319] -> 77(758.47-836.47) [695-1762] -> 65(840.6-918.6) [735-1583] -> 52(924.92-1043.92) [620-1512] ->
    22(1050.63-1155.63) [704-1251] -> D (1172.12)
Nurse 13        123.41  183     D(0) -> 91(134.0-209.0) [134-626] -> 60(272.0-391.0) [272-573] -> 13(400.06-489.06)
    [153-1015] -> 38(501.1-598.1) [42-1194] -> 18(600.93-718.93) [572-1733] -> 25(730.63-819.63) [178-863] ->
    43(827.69-906.69) [621-909] -> 92(914.5-1025.5) [418-1187] -> D (1070.41)
```

```
Objective value (total duration): 1521.14
```

B Test Instance 2 Solution

```
Nurse capacity: 204
Depot return time: 1316
```

```
Nurse 1 126.56  194     D(0) -> 15(211.0-329.0) [211-508] -> 81(344.03-407.03) [206-642] -> 21(411.51-500.51)
    [281-743] -> 37(503.51-602.51) [76-727] -> 9(657.0-754.0) [657-1719] -> 55(771.09-863.09) [50-1211] ->
    67(869.17-958.17) [756-1557] -> 78(974.81-1069.81) [702-1877] -> D (1095.78)
Nurse 2 123.53  182     D(0) -> 12(50.93-147.93) [41-469] -> 90(201.0-275.0) [201-1315] -> 7(292.89-406.89)
    [144-650] -> 53(411.89-525.89) [301-597] -> 94(726.0-837.0) [726-1089] -> 16(859.2-969.2) [599-1171] -> D
    (981.57)
Nurse 3 122.93  198     D(0) -> 85(122.0-216.0) [122-644] -> 79(258.0-358.0) [258-1234] -> 41(555.0-672.0) [555-778]
     -> 18(759.0-849.0) [759-1014] -> 30(853.12-941.12) [563-1704] -> 70(946.12-1045.12) [573-1736] ->
    86(1050.12-1166.12) [782-1827] -> 49(1179.58-1269.58) [646-1319] -> D (1294.82)
Nurse 4 109.8   187     D(0) -> 58(119.0-201.0) [119-550] -> 56(202.41-274.41) [164-324] -> 36(285.18-346.18)
    [158-432] -> 32(357.36-466.36) [225-1333] -> 8(471.46-567.46) [249-1275] -> 6(577.31-655.31) [304-679] ->
    66(663.92-758.92) [506-765] -> 73(764.3-876.3) [54-1226] -> D (899.32)
Nurse 5 137.95  146     D(0) -> 43(52.0-137.0) [52-317] -> 77(579.0-696.0) [579-883] -> 63(787.0-850.0) [787-1689]
   -> 4(866.0-980.0) [725-1131] -> 42(996.76-1108.76) [675-1365] -> 38(1140.97-1226.97) [382-1306] -> D (1233.67)
Nurse 6 155.59  189     D(0) -> 2(51.88-130.88) [38-191] -> 68(140.78-245.78) [128-1323] -> 10(262.78-363.78)
    [2-673] -> 88(537.0-598.0) [537-1408] -> 17(609.0-677.0) [569-1658] -> 20(699.63-816.63) [494-817] ->
```

```
    28(823.91-900.91) [440-1476] -> 40(903.91-991.91) [182-1190] -> D (1016.26)
Nurse 7 129.91  198     D(0) -> 33(19.0-139.0) [19-890] -> 46(269.0-361.0) [269-571] -> 51(373.81-487.81) [217-989]
    -> 82(505.0-572.0) [505-1020] -> 19(579.21-658.21) [145-1078] -> 45(668.21-771.21) [643-1618] ->
    48(802.08-875.08) [581-964] -> D (889.5)
Nurse 8 149.61  193     D(0) -> 74(227.0-332.0) [227-1416] -> 29(347.81-445.81) [255-669] -> 80(452.81-524.81)
    [133-979] -> 91(583.0-669.0) [583-772] -> 75(676.62-781.62) [301-1427] -> 69(791.52-900.52) [299-1033] ->
    92(906.92-967.92) [552-1000] -> 47(973.92-1085.92) [662-1226] -> 93(1114.78-1232.78) [627-1798] -> D (1247.78)
Nurse 9 66.92   177     D(0) -> 72(221.0-323.0) [221-1137] -> 54(422.0-541.0) [422-710] -> 52(617.0-680.0) [617-845]
    -> 39(729.0-847.0) [729-930] -> 60(851.47-920.47) [719-996] -> 14(931.52-1049.52) [241-1360] -> D (1053.64)
Nurse 10        124.85  191     D(0) -> 24(191.0-306.0) [191-635] -> 89(313.81-397.81) [114-675] -> 1(524.0-606.0)
    [524-1236] -> 84(620.42-692.42) [32-1171] -> 71(705.46-776.46) [319-1182] -> 35(782.12-858.12) [266-1010] -> D
    (903.39)
Nurse 11        86.81   177     D(0) -> 62(332.0-396.0) [332-822] -> 31(462.0-552.0) [462-562] -> 87(700.0-810.0)
    [700-961] -> 44(815.39-918.39) [674-1010] -> 5(922.63-1007.63) [196-1294] -> 61(1016.85-1079.85) [737-1917] ->
    83(1083.01-1160.01) [743-1271] -> D (1193.61)
Nurse 12        78.75   187     D(0) -> 23(49.0-121.0) [49-470] -> 25(136.26-234.26) [109-434] -> 59(294.0-378.0)
    [294-540] -> 76(382.12-495.12) [367-699] -> 3(500.22-585.22) [436-1247] -> 65(597.87-707.87) [412-884] ->
    57(716.42-792.42) [651-1159] -> D (801.85)
Nurse 13        174.76  193     D(0) -> 50(183.0-260.0) [183-415] -> 22(288.0-350.0) [288-479] -> 27(424.0-536.0)
    [424-1508] -> 11(540.0-634.0) [26-737] -> 34(663.0-734.0) [663-1322] -> 26(743.06-838.06) [691-1398] ->
    13(838.06-904.06) [757-1753] -> 64(934.86-1030.86) [678-1403] -> D (1070.89)
```

```
Objective value (total duration): 1587.97
```

C Test Instance 3 Solution

```
Nurse capacity: 209
Depot return time: 1475
```

```
Nurse 1 163.24  209     D(0) -> 99(193.0-270.0) [193-709] -> 92(392.0-510.0) [392-526] -> 38(524.04-601.04) [47-646]
    -> 55(604.64-705.64) [354-1340] -> 105(725.29-833.29) [633-1578] -> 44(851.73-943.73) [701-1060] ->
    47(950.13-1047.13) [332-1460] -> 58(1051.25-1114.25) [770-1697] -> D (1144.86)
Nurse 2 60.47   192     D(0) -> 42(103.0-209.0) [103-445] -> 94(216.0-331.0) [186-771] -> 52(708.0-817.0) [708-1627]
```

-> 66(822.0-933.0) [272-1155] -> 8(939.0-1038.0) [648-1424] -> 82(1045.0-1119.0) [471-1441] -> D (1126.07)
Nurse 3 167.53 201 D(0) -> 34(137.0-242.0) [137-492] -> 22(264.47-366.47) [116-416] -> 127(390.07-509.07) [227-586] -> 45(645.0-726.0) [645-1518] -> 106(736.0-849.0) [736-882] -> 101(861.53-933.53) [732-1545] -> 57(948.53-1043.53) [538-1199] -> 120(1044.94-1114.94) [615-1607] -> D (1178.65)
Nurse 4 72.89 199 D(0) -> 53(53.0-146.0) [53-576] -> 129(641.0-727.0) [641-782] -> 5(740.15-831.15) [532-1641] -> 11(836.98-939.98) [282-1259] -> 104(943.15-1018.15) [632-1599] -> 43(1028.44-1131.44) [668-1641] -> 50(1135.44-1236.44) [539-1489] -> D (1255.55)
Nurse 5 175.74 195 D(0) -> 85(48.17-118.17) [8-232] -> 134(130.7-242.7) [56-935] -> 138(259.73-343.73) [134-368] -> 29(462.0-546.0) [462-1057] -> 102(670.0-773.0) [670-1184] -> 12(783.82-895.82) [698-951] -> 83(900.82-973.82) [367-1459] -> 107(979.9-1064.9) [216-1183] -> 86(1074.75-1165.75) [735-1351] -> 31(1205.04-1309.04) [611-1451] -> D (1329.66)
Nurse 6 149.52 199 D(0) -> 9(163.0-251.0) [163-627] -> 13(424.0-487.0) [424-622] -> 132(575.0-678.0) [575-800] -> 19(679.0-785.0) [626-884] -> 108(795.44-865.44) [293-912] -> 119(870.83-969.83) [679-1384] -> 143(988.18-1076.18) [640-1088] -> 74(1086.18-1171.18) [577-1375] -> D (1207.86)
Nurse 7 68.45 183 D(0) -> 111(178.0-276.0) [178-631] -> 63(278.0-340.0) [25-1152] -> 14(361.54-447.54) [275-527] -> 2(607.0-724.0) [607-1148] -> 69(728.47-829.47) [356-1482] -> 135(834.86-899.86) [121-1175] -> 20(905.86-980.86) [178-1117] -> 23(984.98-1060.98) [718-1316] -> D (1065.98)
Nurse 8 21.81 56 D(0) -> 17(421.0-492.0) [421-832] -> 136(737.0-857.0) [737-873] -> D (867.63)
Nurse 9 143.66 205 D(0) -> 137(35.85-135.85) [30-254] -> 46(243.0-314.0) [243-869] -> 125(322.06-441.06) [42-623] -> 91(484.0-567.0) [484-589] -> 73(572.0-643.0) [183-842] -> 21(645.83-706.83) [369-1228] -> 103(709.06-799.06) [680-1781] -> 15(814.1-878.1) [539-1298] -> 67(898.02-977.02) [225-1051] -> D (1010.96)
Nurse 10 86.01 200 D(0) -> 87(277.0-362.0) [277-779] -> 97(383.02-450.02) [293-1368] -> 118(458.27-565.27) [302-1086] -> 113(566.68-633.68) [382-699] -> 68(640.09-735.09) [494-1332] -> 41(744.31-835.31) [707-1189] -> 124(845.36-922.36) [660-1773] -> D (936.39)
Nurse 11 125.94 200 D(0) -> 48(296.0-387.0) [296-713] -> 39(416.73-503.73) [293-540] -> 35(508.73-583.73) [247-1067] -> 78(585.73-664.73) [304-1111] -> 93(666.15-739.15) [278-1040] -> 112(755.18-831.18) [735-1293] -> 25(848.29-935.29) [733-1427] -> D (958.9)
Nurse 12 110.18 186 D(0) -> 114(270.0-340.0) [270-421] -> 59(514.0-616.0) [514-1589] -> 95(629.93-718.93) [605-1375] -> 140(782.0-894.0) [782-943] -> 64(899.39-968.39) [534-1490] -> 115(972.51-1092.51) [394-1228] -> 81(1102.71-1174.71) [478-1450] -> D (1195.73)
Nurse 13 124.15 187 D(0) -> 16(91.0-167.0) [91-823] -> 1(190.32-289.32) [105-359] -> 65(298.54-392.54) [51-1196] -> 75(466.0-564.0) [466-583] -> 54(570.71-670.71) [471-911] -> 100(673.87-761.87) [634-830] -> 18(769.87-873.87) [257-1359] -> 126(880.2-975.2) [445-1529] -> D (1004.35)
Nurse 14 120.5 208 D(0) -> 90(62.0-128.0) [62-1046] -> 56(132.24-202.24) [12-310] -> 142(213.29-292.29) [211-1312] -> 79(296.53-404.53) [289-451] -> 62(417.68-489.68) [183-633] -> 121(499.88-601.88) [465-1456] ->

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    128(635.0-717.0) [635-737] -> 84(719.24-805.24) [668-1521] -> 30(819.66-892.66) [145-1332] -> D (914.13)
Nurse 15        147.89  180     D(0) -> 123(132.0-196.0) [132-707] -> 71(198.0-271.0) [93-1269] -> 10(284.04-403.04)
     [222-1122] -> 3(414.08-481.08) [76-1150] -> 40(495.51-604.51) [473-877] -> 89(610.83-688.83) [28-1182] ->
    24(704.64-766.64) [615-1293] -> 4(770.77-832.77) [764-1094] -> 139(852.86-958.86) [725-1005] -> D (992.11)
Nurse 16        111.91  200     D(0) -> 130(111.0-210.0) [111-384] -> 32(216.71-324.71) [93-1044] -> 88(373.0-491.0)
     [373-496] -> 77(584.0-680.0) [584-1379] -> 26(795.0-910.0) [795-1047] -> 122(911.41-1018.41) [273-1226] ->
    51(1026.22-1142.22) [603-1538] -> 7(1149.84-1211.84) [746-1636] -> D (1242.11)
Nurse 17        146.53  196     D(0) -> 133(206.0-309.0) [206-1340] -> 70(325.76-385.76) [124-683] -> 6(459.0-564.0)
     [459-782] -> 61(566.0-647.0) [479-1468] -> 109(740.0-846.0) [740-861] -> 117(857.31-939.31) [136-1161] ->
    131(957.12-1069.12) [406-1153] -> D (1108.56)
Nurse 18        158.34  203     D(0) -> 110(267.0-360.0) [267-869] -> 60(373.42-445.42) [330-668] -> 72(529.0-637.0)
     [529-1661] -> 98(638.0-715.0) [597-1441] -> 37(727.17-806.17) [251-1331] -> 141(817.57-905.57) [756-969] ->
    28(911.89-1012.89) [36-1153] -> 80(1025.06-1134.06) [769-1736] -> D (1180.59)
Nurse 19        134.38  205     D(0) -> 27(126.0-216.0) [126-280] -> 49(274.0-356.0) [274-1259] -> 36(533.0-647.0)
    [533-1450] -> 96(655.25-748.25) [609-1001] -> 33(755.86-818.86) [731-1591] -> 76(830.9-940.9) [530-1119] ->
    116(952.08-1029.08) [181-1360] -> D (1064.55)
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Objective value (total duration): 2289.15
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